

GPU 설정 및 드라이버, CUDA, cuDNN 설치 가이드



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GPU 설정 및 드라이버, CUDA, cuDNN 설치 가이드

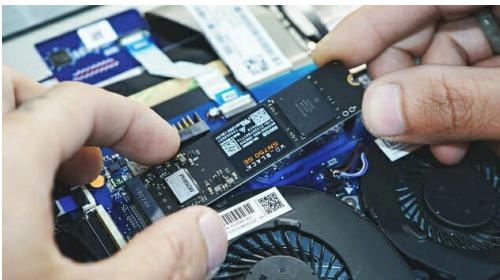
Exported from Confluence · <https://pinkwink.atlassian.net>

GPU 설정

중요한 건 나의 장비를 내가 스스로 세팅할 수 있어야 한다는 것~

또 하나는 포맷 및 재설정을 두려워 말것~ [좀 짜증은 나겠지만~]

PCI 장비라는 것이 있다. 데스크탑도, 노트북도



내 PC에 PCI 장치 중에 그래픽 카드 관련 목록을 확인해보면~

```
sohee@pinklab:~$ lspci | grep VGA
0000:00:02.0 VGA compatible controller: Intel Corporation TigerLake-LP GT2 [Iris Xe Graphics] (rev 01)
0000:01:00.0 VGA compatible controller: NVIDIA Corporation GA106M [GeForce RTX 3060 Mobile / Max-Q] (rev a1)
```

NVIDIA 그래픽 카드가 있는 노트북들은 대체로 그래픽카드를 두 개 가지고 있다.

더 상세한 정보는

```

pinklab@pc:~$ sudo lshw -C display
*-display
    description: VGA compatible controller
    product: AD107M [GeForce RTX 4060 Max-Q / Mobile]
    vendor: NVIDIA Corporation
    physical id: 0
    bus info: pci@0000:01:00.0
    logical name: /dev/fb0
    version: a1
    width: 64 bits
    clock: 33MHz
    capabilities: pm msi pciexpress vga_controller bus_master cap_list rom fb
    configuration: depth=32 driver=nvidia latency=0 mode=1920x1080 visual=truecolor xres=19
20 yres=1080
    resources: iomemory:600-5ff iomemory:620-61f irq:191 memory:78000000-78ffffff memory:60
00000000-61ffffffff memory:6200000000-6201ffffff ioport:4000(size=128) memory:79000000-7907fff
f
*-display
    description: VGA compatible controller
    product: Raptor Lake-P [UHD Graphics]
    vendor: Intel Corporation
    physical id: 2
    bus info: pci@0000:00:02.0
    logical name: /dev/fb0
    version: 04
    width: 64 bits
    clock: 33MHz
    capabilities: pciexpress msi pm vga_controller bus_master cap_list rom fb
    configuration: depth=32 driver=i915 latency=0 resolution=1920,1080
    resources: iomemory:620-61f iomemory:400-3ff irq:190 memory:6204000000-6204ffffff mem
ory:4000000000-400ffffff ioport:5000(size=64) memory:c0000-dffff memory:4010000000-4016ffffff m
emory:4020000000-40ffffff

```

일단 중요한 설치를 앞두고 있으므로

```

sohee@pinklab:~$ sudo apt update && sudo apt upgrade -y
Get:1 file:/var/cudnn-local-repo-ubuntu2404-9.16.0 InRelease [1,572 B]
Get:1 file:/var/cudnn-local-repo-ubuntu2404-9.16.0 InRelease [1,572 B]
Hit:2 https://dl.winehq.org/wine-builds/ubuntu noble InRelease
Hit:3 http://mirror.kakao.com/ubuntu noble InRelease
Hit:4 http://mirror.kakao.com/ubuntu noble-updates InRelease
Hit:5 http://mirror.kakao.com/ubuntu noble-backports InRelease
Hit:6 https://dl.google.com/linux/chrome/deb stable InRelease
Hit:7 http://security.ubuntu.com/ubuntu xenial-security InRelease
Hit:8 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2404/x86_64 InRelease
Hit:9 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:11 https://ppa.launchpadcontent.net/graphics-drivers/ppa/ubuntu noble InRelease
Hit:10 https://packagecloud.io/slacktechnologies/slack/debian jessie InRelease

```

먼저 드라이버가 설치되어 있는지 확인

```
sohee@pinklab:~$ nvidia-smi
```

Wed Dec 17 16:48:05 2025

NVIDIA-SMI 580.105.08				Driver Version: 580.105.08				CUDA Version: 13.0			
+-----+											

정상적으로 실행이 안된다면 드라이버 설치가 필요합니다.

```
addinedu@pc:~$ nvidia-smi
```

Command 'nvidia-smi' not found, but can be installed with:

```
sudo apt install nvidia-utils-470 # version 470.256.02-0ubuntu0.24.04.1, or
sudo apt install nvidia-utils-470-server # version 470.256.02-0ubuntu0.24.04.1
sudo apt install nvidia-utils-535 # version 535.274.02-0ubuntu0.24.04.2
sudo apt install nvidia-utils-535-server # version 535.274.02-0ubuntu0.24.04.2
sudo apt install nvidia-utils-570 # version 570.195.03-0ubuntu0.24.04.1
sudo apt install nvidia-utils-525 # version 525.147.05-0ubuntu1
sudo apt install nvidia-utils-525-server # version 525.147.05-0ubuntu1
sudo apt install nvidia-utils-565-server # version 565.57.01-0ubuntu0.24.04.3
sudo apt install nvidia-utils-570-server # version 570.195.03-0ubuntu0.24.04.3
sudo apt install nvidia-utils-580 # version 580.95.05-0ubuntu0.24.04.3
sudo apt install nvidia-utils-580-server # version 580.95.05-0ubuntu0.24.04.3
sudo apt install nvidia-utils-550-server # version 550.163.01-0ubuntu0.24.04.1
```

드라이버 설치

그래픽 카드 정보 확인

- 내 그래픽 카드와 호환 되는 드라이버들을 보여준다

```
sohee@pinklab:~$ ubuntu-drivers devices
== /sys/devices/pci0000:00/0000:00:06.0/0000:01:00.0 ==
modalias : pci:v000010DEd00002520sv00001043sd000016E2bc03sc00i00
vendor    : NVIDIA Corporation
model     : GA106M [GeForce RTX 3060 Mobile / Max-Q]
driver    : nvidia-driver-535-server-open - distro non-free
driver    : nvidia-driver-470-server - distro non-free
driver    : nvidia-driver-565-open - third-party non-free
driver    : nvidia-driver-570-server - distro non-free
driver    : nvidia-driver-575 - third-party non-free
driver    : nvidia-driver-550-open - third-party non-free
driver    : nvidia-driver-570-server-open - distro non-free
driver    : nvidia-driver-550 - third-party non-free
driver    : nvidia-driver-535-open - distro non-free
driver    : nvidia-driver-570 - third-party non-free
driver    : nvidia-driver-470 - third-party non-free
driver    : nvidia-driver-570-open - third-party non-free
driver    : nvidia-driver-580-open - third-party non-free recommended
driver    : nvidia-driver-555 - third-party non-free
driver    : nvidia-driver-580-server-open - distro non-free
driver    : nvidia-driver-560 - third-party non-free
driver    : nvidia-driver-555-open - third-party non-free
driver    : nvidia-driver-535-server - distro non-free
driver    : nvidia-driver-580 - third-party non-free
driver    : nvidia-driver-560-open - third-party non-free
driver    : nvidia-driver-565 - third-party non-free
driver    : nvidia-driver-575-open - third-party non-free
driver    : nvidia-driver-580-server - distro non-free
driver    : nvidia-driver-535 - distro non-free
driver    : xserver-xorg-video-nouveau - distro free builtin
```

그래픽 카드 드라이버 설치

- 위에서 확인한 목록 중 'recommended' 항목을 선택하는 것을 추천

```
sohee@pinklab:~$ sudo apt install nvidia-driver-580-open
```

재부팅

```
sohee@pinklab:~$ reboot
```

(Optional) 만약 오류가 발생했다면 NVIDIA 드라이버를 삭제 후 재설치

bash

```
1 # 기존에 설치된 NVIDIA Driver 삭제
2 sudo apt-get --purge -y remove *nvidia*
3 sudo apt-get autoremove
4 sudo apt-get autoclean
5
6 # 삭제되었는지 확인
7 sudo dpkg -l | grep nvidia
```

CUDA 설치

나의 환경에 맞는 CUDA, cuDNN 버전 확인

<https://docs.nvidia.com/deeplearning/cudnn/backend/latest/reference/support-matrix.html>

(<https://docs.nvidia.com/deeplearning/cudnn/backend/latest/reference/support-matrix.html>)

Supported NVIDIA Hardware and CUDA Version

cuDNN Package ^[1]	CUDA Toolkit Version	Supports static linking? ^[2]	NVIDIA Driver Version for Linux	NVIDIA Driver Version for Windows	CUDA Compute Capability	Supported NVIDIA GPU Architectures
cuDNN 9.17.0 for CUDA 13.x	13.1 13.0	Yes	$\geq 580.65.06$	N/A	12.0 10.0 9.0 8.9 8.6 8.0 7.5	- NVIDIA Blackwell - NVIDIA Hopper - NVIDIA Ada Lovelace - NVIDIA Ampere - NVIDIA Turing
cuDNN 9.17.0 for CUDA 12.x	12.9 12.8 12.6 12.5 12.4 12.3 12.2 12.1 12.0	Yes	$\geq 525.60.13$	≥ 527.41	12.0 ^[3] 10.0 ^[3] 9.0 8.9 8.6 8.0 7.5	- NVIDIA Blackwell ^[3] - NVIDIA Hopper - NVIDIA Ada Lovelace - NVIDIA Ampere - NVIDIA Turing

CUDA Toolkit Archive

<https://developer.nvidia.com/cuda-toolkit-archive> (<https://developer.nvidia.com/cuda-toolkit-archive>)

원하는 CUDA 버전을 선택

CUDA Toolkit Archive

Previous releases of the CUDA Toolkit, GPU Computing SDK, documentation and developer drivers can be found using the links below. Please select the release you want from the list below, and be sure to check www.nvidia.com/drivers for more recent production drivers appropriate for your hardware configuration.

[Download Latest CUDA Toolkit](#)

[Learn More about CUDA Toolkit](#)

Latest Release

CUDA Toolkit 13.1.0 (December 2025), [Versioned Online Documentation](#)

Archived Releases

[CUDA Toolkit 13.0.2](#) (October 2025), [Versioned Online Documentation](#)

[CUDA Toolkit 13.0.1](#) (September 2025), [Versioned Online Documentation](#)

[CUDA Toolkit 13.0.0](#) (August 2025), [Versioned Online Documentation](#)

[CUDA Toolkit 12.9.1](#) (June 2025), [Versioned Online Documentation](#)

[CUDA Toolkit 12.9.0](#) (May 2025), [Versioned Online Documentation](#)

[CUDA Toolkit 12.8.1](#) (March 2025), [Versioned Online Documentation](#)

[CUDA Toolkit 12.8.0](#) (January 2025), [Versioned Online Documentation](#)

본인의 환경에 맞는 옵션을 선택

CUDA Toolkit 13.1 Downloads

Select Target Platform

Click on the green buttons that describe your target platform. Only supported platforms will be shown. By downloading and using the software, you agree to fully comply with the terms and conditions of the [CUDA EULA](#).

Operating System	Linux	Windows					
Architecture	x86_64	arm64-sbsa					
Distribution	Amazon-Linux	Azure-Linux	Debian	Fedora	KylinOS	OpenSUSE	Oracle-Linux
	RHEL	Rocky	SLES	Ubuntu	WSL-Ubuntu		
Version	22.04	24.04					
Installer Type	deb (local)	deb (network)	runfile (local)				

옵션을 모두 선택하면 설치 방법이 등장

Download Installer for Linux Ubuntu 24.04 x86_64

The base installer is available for download below.

> CUDA Toolkit Installer

Installation Instructions:

```
$ wget https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2404/x86_64/cuda-keyring_1.1-1_all.deb
$ sudo dpkg -i cuda-keyring_1.1-1_all.deb
$ sudo apt-get update
$ sudo apt-get -y install cuda-toolkit-13-1
```

Additional installation options are detailed [here](#).

> Driver Installer

NVIDIA Driver Instructions (choose one option)

To install the open kernel module flavor:

```
$ sudo apt-get install -y nvidia-open
```

To install the proprietary kernel module flavor:

```
$ sudo apt-get install -y cuda-drivers
```

NVIDIA CUDA 저장소 등록용 패키지를 다운로드

Ubuntu가 NVIDIA CUDA 저장소를 신뢰하고 사용할 수 있도록 해주는 저장소 키링 패키지

```
sohee@pinklab:~$ cd Downloads/
sohee@pinklab:~/Downloads$ wget https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2404/x86_64/cuda-keyring_1.1-1_all.deb
--2025-12-17 17:49:32-- https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2404/x86_64/cuda-keyring_1.1-1_all.deb
Resolving developer.download.nvidia.com (developer.download.nvidia.com)... 104.103.68.177, 104.103.68.146, 104.103.68.154, ...
Connecting to developer.download.nvidia.com (developer.download.nvidia.com)|104.103.68.177|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 4328 (4.2K) [application/x-deb]
Saving to: 'cuda-keyring_1.1-1_all.deb.1'

cuda-keyring_1.1-1_al 100%[=====>] 4.23K --.-KB/s in 0s

2025-12-17 17:49:33 (496 MB/s) - 'cuda-keyring_1.1-1_all.deb.1' saved [4328/4328]
```

방금 다운로드한 cuda-keyring 패키지를 설치

Ubuntu 시스템에 NVIDIA CUDA 저장소 정보와 인증 키가 등록

```
sohee@pinklab:~/Downloads$ sudo dpkg -i cuda-keyring_1.1-1_all.deb
Selecting previously unselected package cuda-keyring.
(Reading database ... 289365 files and directories currently installed.)
Preparing to unpack cuda-keyring_1.1-1_all.deb ...
Unpacking cuda-keyring (1.1-1) ...
Setting up cuda-keyring (1.1-1) ...
```

APT 패키지 목록을 새로고침

새로 등록된 NVIDIA CUDA 저장소에서 설치 가능한 패키지 목록을 다시 읽어옴

```
sohee@pinklab:~/Downloads$ sudo apt-get update
Hit:1 http://mirror.kakao.com/ubuntu noble InRelease
Hit:2 http://mirror.kakao.com/ubuntu noble-updates InRelease
Hit:3 http://mirror.kakao.com/ubuntu noble-backports InRelease
Hit:4 https://dl.google.com/linux/chrome/deb stable InRelease
Hit:5 https://dl.winehq.org/wine-builds/ubuntu noble InRelease
Hit:6 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:7 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2404/x86_64 InRelease
Hit:9 https://ppa.launchpadcontent.net/graphics-drivers/ppa/ubuntu noble InRelease
Hit:8 https://packagecloud.io/slacktechnologies/slack/debian jessie InRelease
Reading package lists... Done
```

CUDA Toolkit 13.1을 설치

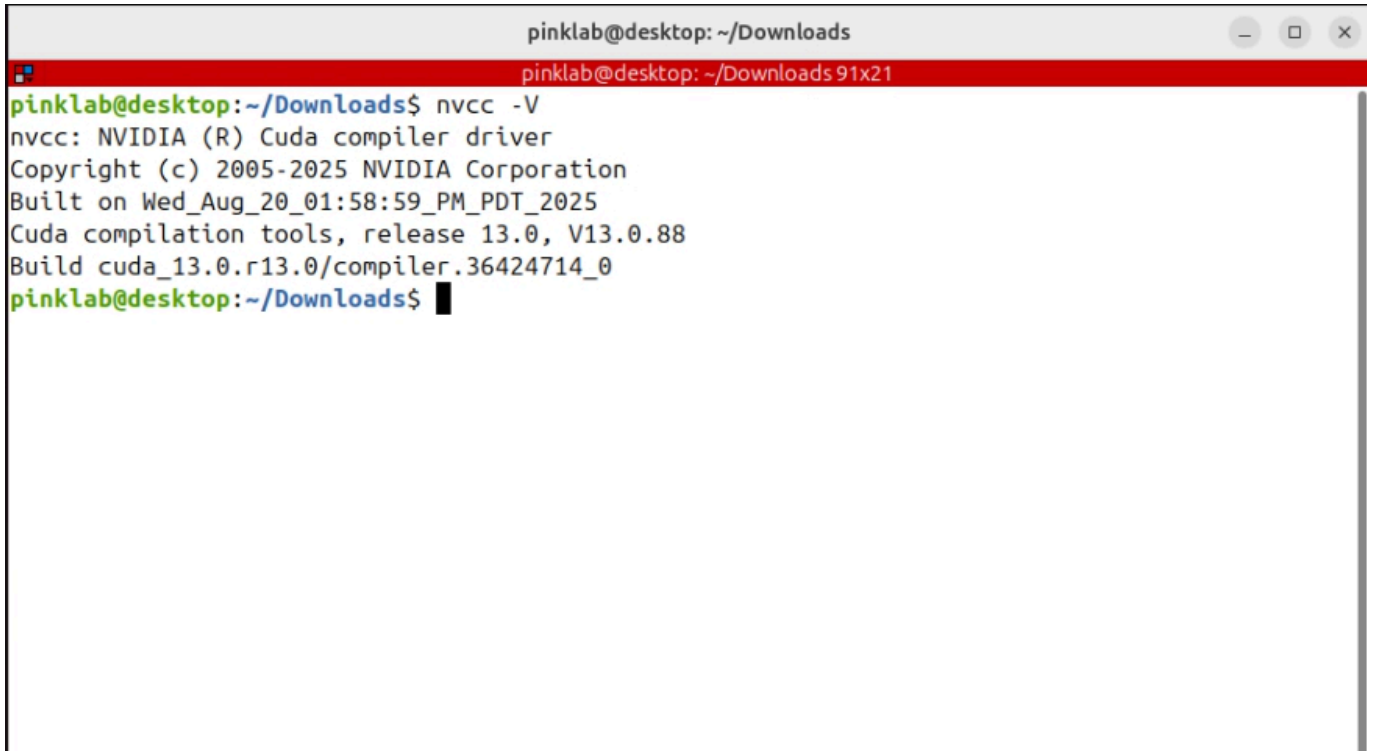
```
sohee@pinklab:~/Downloads$ sudo apt-get -y install cuda-toolkit-13-1
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  cuda-cccl-13-1 cuda-command-line-tools-13-1 cuda-compiler-13-1 cuda-crt-13-1
  cuda-cudart-13-1 cuda-cudart-dev-13-1 cuda-culibos-dev-13-1 cuda-cuobjdump-13-1
  cuda-cupti-13-1 cuda-cupti-dev-13-1 cuda-cuxxfilt-13-1 cuda-documentation-13-1
  cuda-driver-dev-13-1 cuda-gdb-13-1 cuda-libraries-13-1 cuda-libraries-dev-13-1
  cuda-nsight-13-1 cuda-nsight-compute-13-1 cuda-nsight-systems-13-1 cuda-nvcc-13-1
  cuda-nvdisasm-13-1 cuda-nvml-dev-13-1 cuda-nvprune-13-1 cuda-nvrtc-13-1
  cuda-nvrtc-dev-13-1 cuda-nvtx-13-1 cuda-opencl-13-1 cuda-profiler-api-13-1
  cuda-sandbox-dev-13-1 cuda-sanitizer-13-1 cuda-tileiras-13-1
  cuda-toolkit-13-1-config-common cuda-toolkit-13-config-common
  cuda-toolkit-config-common cuda-tools-13-1 cuda-visual-tools-13-1 gds-tools-13-1
  libcublas-13-1 libcublas-dev-13-1 libcufft-13-1 libcufft-dev-13-1 libcufile-13-1
  libcufile-dev-13-1 libcurand-13-1 libcurand-dev-13-1 libcusolver-13-1
  libcusolver-dev-13-1 libcusparse-13-1 libcusparse-dev-13-1 libnpp-13-1
  libnpp-dev-13-1 libnvfatbin-13-1 libnvfatbin-dev-13-1 libnvjitlink-13-1
  libnvjitlink-dev-13-1 libnvjpeg-13-1 libnvjpeg-dev-13-1 libnvptxcompiler-13-1
  libnvvm-13-1 libxcb-cursor0 nsight-compute-2025.4.0 nsight-systems-2025.5.2
```

설치 후에 경로 잡아주기

Code

```
1 echo 'export PATH=/usr/local/cuda-13.1/bin:$PATH' >> ~/.bashrc
2 echo 'export LD_LIBRARY_PATH=/usr/local/cuda-13.1/lib64${LD_LIBRARY_PATH:+:${LD_LIBR
  ARY_PATH}}' >> ~/.bashrc
3 source ~/.bashrc
```

CUDA 버전 확인

A terminal window titled 'pinklab@desktop: ~/Downloads' with standard window controls. The terminal shows the command 'nvcc -V' being executed. The output text is: 'nvcc: NVIDIA (R) Cuda compiler driver', 'Copyright (c) 2005-2025 NVIDIA Corporation', 'Built on Wed_Aug_20_01:58:59_PM_PDT_2025', 'Cuda compilation tools, release 13.0, V13.0.88', and 'Build cuda_13.0.r13.0/compiler.36424714_0'. The prompt returns to 'pinklab@desktop:~/Downloads\$'.

```
pinklab@desktop: ~/Downloads
pinklab@desktop: ~/Downloads 91x21
pinklab@desktop:~/Downloads$ nvcc -V
nvcc: NVIDIA (R) Cuda compiler driver
Copyright (c) 2005-2025 NVIDIA Corporation
Built on Wed_Aug_20_01:58:59_PM_PDT_2025
Cuda compilation tools, release 13.0, V13.0.88
Build cuda_13.0.r13.0/compiler.36424714_0
pinklab@desktop:~/Downloads$
```

cuDNN 설치

<https://developer.nvidia.com/cudnn-archive> (<https://developer.nvidia.com/cudnn-archive>)

나의 환경에 맞는 cuDNN 설치

cuDNN Archive

Latest Release

[cuDNN 9.17.0 \(December 2025\), Documentation](#)

Archived Releases

- [cuDNN 9.16.0 \(November 2025\)](#)
- [cuDNN 9.15.1 \(November 2025\)](#)
- [cuDNN 9.15.0 \(November 2025\)](#)
- [cuDNN 9.14.0 \(October 2025\)](#)
- [cuDNN 9.13.1 \(September 2025\)](#)
- [cuDNN 9.13.0 \(September 2025\)](#)

나의 환경에 맞게 선택 하면 설치 방법이 나타남

cuDNN 9.17.0 Downloads

Select Target Platform

Click on the green buttons that describe your target platform. Only supported platforms will be shown. By downloading and using the software, you agree to fully comply with the terms and conditions of the [NVIDIA Software License Agreement](#).

Operating System	Linux Windows
Architecture	x86_64 arm64-sbsa aarch64-jetson
Distribution	Tarball Debian OpenSUSE RHEL Rocky SLES Ubuntu
Version	22.04 24.04
Installer Type	deb (local) deb (network)
Configuration	FULL JIT

설명대로 설치 진행

Download Installer for Linux Ubuntu 24.04 x86_64

The base installer is available for download below.

> Base Installer

Download (1.5 GB)

Installation Instructions:

```
$ wget https://developer.download.nvidia.com/compute/cudnn/9.17.0/local_installers/cudnn-local-repo-ubuntu2404-9.17.0_1.0-1_amd64.deb
$ sudo dpkg -i cudnn-local-repo-ubuntu2404-9.17.0_1.0-1_amd64.deb
$ sudo cp /var/cudnn-local-repo-ubuntu2404-9.17.0/cudnn-*-keyring.gpg /usr/share/keyrings/
$ sudo apt-get update
$ sudo apt-get -y install cudnn
```

For more details about cuDNN JIT and how it compares to cuDNN FULL, please see [here](#).

To install for CUDA 12, perform the above configuration but install the CUDA 12 specific package:

```
$ sudo apt-get -y install cudnn9-cuda-12
```

To install for CUDA 13, perform the above configuration but install the CUDA 13 specific package:

```
$ sudo apt-get -y install cudnn9-cuda-13
```

cuDNN 로컬 저장소 .deb 파일 다운로드

NVIDIA 서버에서 cuDNN 설치용 로컬 저장소 패키지를 다운로드

```
sohee@pinklab:~/Downloads$ wget https://developer.download.nvidia.com/compute/cudnn/9.17.0/local_installers/cudnn-local-repo-ubuntu2404-9.17.0_1.0-1_amd64.deb
--2025-12-18 09:35:56-- https://developer.download.nvidia.com/compute/cudnn/9.17.0/local_installers/cudnn-local-repo-ubuntu2404-9.17.0_1.0-1_amd64.deb
Resolving developer.download.nvidia.com (developer.download.nvidia.com)... 2.16.244.11, 2.16.244.17
Connecting to developer.download.nvidia.com (developer.download.nvidia.com)|2.16.244.11|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1570453622 (1.5G) [application/x-deb]
Saving to: 'cudnn-local-repo-ubuntu2404-9.17.0_1.0-1_amd64.deb'
```

```
cudnn-local-repo-ubun 100%[=====>] 1.46G 16.3MB/s in 2m 18s
```

```
2025-12-18 09:38:16 (10.9 MB/s) - 'cudnn-local-repo-ubuntu2404-9.17.0_1.0-1_amd64.deb'
saved [1570453622/1570453622]
```

다운로드한 .deb 패키지 설치


```
sohee@pinklab:~/Downloads$ sudo dpkg -i cudnn-local-repo-ubuntu2404-9.17.0_1.0-1_amd64.deb
Selecting previously unselected package cudnn-local-repo-ubuntu2404-9.17.0.
(Reading database ... 306235 files and directories currently installed.)
Preparing to unpack cudnn-local-repo-ubuntu2404-9.17.0_1.0-1_amd64.deb ...
Unpacking cudnn-local-repo-ubuntu2404-9.17.0 (1.0-1) ...
Setting up cudnn-local-repo-ubuntu2404-9.17.0 (1.0-1) ...
```

The public cudnn-local-repo-ubuntu2404-9.17.0 GPG key does not appear to be installed.
To install the key, run this command:
sudo cp /var/cudnn-local-repo-ubuntu2404-9.17.0/cudnn-local-1A5BEAD5-keyring.gpg /usr/share/keyrings/

NVIDIA 저장소 인증 키를 시스템 키링 위치로 복사

APT가 이 cuDNN 저장소를 신뢰할 수 있도록 인증 키를 등록

```
sohee@pinklab:~/Downloads$ sudo cp /var/cudnn-local-repo-ubuntu2404-9.17.0/
cudnn-*-keyring.gpg /usr/share/keyrings/
```

APT 패키지 목록 갱신

방금 추가한 cuDNN 로컬 저장소의 패키지 목록을 Ubuntu가 다시 읽도록 해준다.

```
sohee@pinklab:~/Downloads$ sudo apt-get update
Get:1 file:/var/cudnn-local-repo-ubuntu2404-9.17.0 InRelease [1,572 B]
Get:1 file:/var/cudnn-local-repo-ubuntu2404-9.17.0 InRelease [1,572 B]
Hit:2 http://mirror.kakao.com/ubuntu noble InRelease
Hit:3 http://mirror.kakao.com/ubuntu noble-updates InRelease
Hit:4 http://mirror.kakao.com/ubuntu noble-backports InRelease
Get:5 file:/var/cudnn-local-repo-ubuntu2404-9.17.0 Packages [5,615 B]
Hit:6 https://dl.winehq.org/wine-builds/ubuntu noble InRelease
Hit:7 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:8 https://dl.google.com/linux/chrome/deb stable InRelease
Hit:10 https://ppa.launchpadcontent.net/graphics-drivers/ppa/ubuntu noble I
nRelease
Hit:9 https://packagecloud.io/slacktechnologies/slack/debian jessie InRelea
se
Hit:11 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2404/
x86_64 InRelease
Reading package lists... Done
```

cuDNN 설치

```
sohee@pinklab:~/Downloads$ sudo apt-get -y install cudnn
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  cudnn9 cudnn9-cuda-13 cudnn9-cuda-13-1 libcudnn9-cuda-13
  libcudnn9-dev-cuda-13 libcudnn9-headers-cuda-13 libcudnn9-samples
  libcudnn9-static-cuda-13
The following NEW packages will be installed:
  cudnn cudnn9 cudnn9-cuda-13 cudnn9-cuda-13-1 libcudnn9-cuda-13
  libcudnn9-dev-cuda-13 libcudnn9-headers-cuda-13 libcudnn9-samples
  libcudnn9-static-cuda-13
0 upgraded, 9 newly installed, 0 to remove and 37 not upgraded.
Need to get 662 MB of archives.
After this operation, 1,167 MB of additional disk space will be used.
```

(선택 사항) 혹시 프로젝트 중에 환경이 망가지는 게 걱정된다면

특정 패키지의 업그레이드를 막을 수 있다.

```
sohee@pinklab:~/Downloads$ sudo nano /etc/apt/apt.conf.d/50unattended-upgrades
```

항목 추가

```

GNU nano 7.2 /etc/apt/apt.conf.d/50unattended-upgrades *
// Automatically upgrade packages from these (origin:archive) pairs
//
// Note that in Ubuntu security updates may pull in new dependencies
// from non-security sources (e.g. chromium). By allowing the release
// pocket these get automatically pulled in.
Unattended-Upgrade::Allowed-Origins {
    "${distro_id}:${distro_codename}";
    "${distro_id}:${distro_codename}-security";
    // Extended Security Maintenance; doesn't necessarily exist for
    // every release and this system may not have it installed, but if
    // available, the policy for updates is such that unattended-upgrades
    // should also install from here by default.
    "${distro_id}ESMA:${distro_codename}-apps-security";
    "${distro_id}ESM:${distro_codename}-infra-security";
//    "${distro_id}:${distro_codename}-updates";
//    "${distro_id}:${distro_codename}-proposed";
//    "${distro_id}:${distro_codename}-backports";
};

// Python regular expressions, matching packages to exclude from upgrading
Unattended-Upgrade::Package-Blacklist {
    "nvidia-";
    "libnvidia-";
    // The following matches all packages starting with linux-
    // "linux-";

    // Use $ to explicitly define the end of a package name. Without
    // the $, "libc6" would match all of them.
    // "libc6$";
    // "libc6-dev$";
    // "libc6-i686$";

    // Special characters need escaping
    // "libstdc\+\+6$";

    // The following matches packages like xen-system-amd64, xen-utils-4.1,
    // xenstore-utils and libxenstore3.0
    // "(lib)?xen(store)?";

    // For more information about Python regular expressions, see
    // https://docs.python.org/3/howto/regex.html
};

```

저장 - Ctrl + X

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute	^C Location	M-U Undo
^X Exit	^R Read File	^L Replace	^U Paste	^J Justify	^/_ Go To Line	M-E Redo

저장 - Y

```

Save modified buffer?
Y Yes
N No          ^C Cancel

```

저장 - 엔터

```
File Name to Write: /etc/apt/apt.conf.d/50unattended-upgrades
^G Help          M-D DOS Format   M-A Append      M-B Backup File
^C Cancel        M-M Mac Format   M-P Prepend     ^T Browse
```

혹시 CUDA 재설치를 위해 삭제가 필요하다면

bash

```
1 # 기존에 설치된 CUDA 삭제
2 sudo rm -rf /usr/local/cuda*
3 sudo apt-get --purge -y remove '*cuda*'
4 sudo apt-get autoremove --purge cuda
5
6 # 삭제 되었는지 확인
7 sudo dpkg -l | grep cuda
```

CUDA 사용 테스트

가상환경 생성 및 활성화

```
pinklab@pc:~$ python3 -m venv ~/venv/tensorflow
pinklab@pc:~$ source ~/venv/tensorflow/bin/activate
```

pip 업그레이드

```
(tensorflow) pinklab@pc:~$ pip install --upgrade pip
Requirement already satisfied: pip in ./venv/tensorflow/lib/python3.12/site-packages (24.0)
Collecting pip
  Using cached pip-25.3-py3-none-any.whl.metadata (4.7 kB)
Using cached pip-25.3-py3-none-any.whl (1.8 MB)
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 24.0
    Uninstalling pip-24.0:
      Successfully uninstalled pip-24.0
Successfully installed pip-25.3
```

Tensorflow 설치

```
(tensorflow) pinklab@pc:~$ python3 -m pip install tensorflow
Collecting tensorflow
  Using cached tensorflow-2.20.0-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (4.5 kB)
Collecting absl-py>=1.0.0 (from tensorflow)
  Using cached absl_py-2.3.1-py3-none-any.whl.metadata (3.3 kB)
Collecting astunparse>=1.6.0 (from tensorflow)
  Using cached astunparse-1.6.3-py2.py3-none-any.whl.metadata (4.4 kB)
Collecting flatbuffers>=24.3.25 (from tensorflow)
  Using cached flatbuffers-25.12.19-py2.py3-none-any.whl.metadata (1.0 kB)
Collecting gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 (from tensorflow)
  Using cached gast-0.7.0-py3-none-any.whl.metadata (1.5 kB)

Installing collected packages: namex, libclang, flatbuffers, wrapt, wheel, urllib3, typing_extensions, termcolor, tensorboard-data-server, six, setuptools, pygments, protobuf, pillow, packaging, opt_einsum, numpy, mdurl, markupsafe, markdown, idna, gast, charset_normalizer, certifi, absl-py, werkzeug, requests, optree, ml_dtypes, markdown-it-py, h5py, grpcio, google_pasta, astunparse, tensorboard, rich, keras, tensorflow
Successfully installed absl-py-2.3.1 astunparse-1.6.3 certifi-2025.11.12 charset_normalizer-3.4.4 flatbuffers-25.12.19 gast-0.7.0 google_pasta-0.2.0 grpcio-1.76.0 h5py-3.15.1 idna-3.11 keras-3.13.0 libclang-18.1.1 markdown-3.10 markdown-it-py-4.0.0 markupsafe-3.0.3 mdurl-0.1.2 ml_dtypes-0.5.4 namex-0.1.0 numpy-2.4.0 opt_einsum-3.4.0 optree-0.18.0 packaging-25.0 pillow-12.0.0 protobuf-6.33.2 pygments-2.19.2 requests-2.32.5 rich-14.2.0 setuptools-80.9.0 six-1.17.0 tensorboard-2.20.0 tensorboard-data-server-0.7.2 tensorflow-2.20.0 termcolor-3.3.0 typing_extensions-4.15.0 urllib3-2.6.2 werkzeug-3.1.4 wheel-0.45.1 wrapt-2.0.1
```

GPU 지원 확인

```
(tensorflow) pinklab@pc:~$ python3
Python 3.12.3 (main, Nov 6 2025, 13:44:16) [GCC 13.3.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> import tensorflow as tf
>>> print(tf.config.list_physical_devices('GPU'))
[PhysicalDevice(name='/physical_device:GPU:0', device_type='GPU')]
```



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